Avalon-MM Firewall IP Core

Block Diagrams and Descriptions

IP core	`altera_avalon_mm_firewall`
Core version	1.0
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Date	August 2026
Target	Intel FPGA / Quartus Prime, Platform Designer

Burst-capable access-control and fault-isolation IP core for Avalon-MM. Source-available at github.com/monkstein88/altera-ip-cores.

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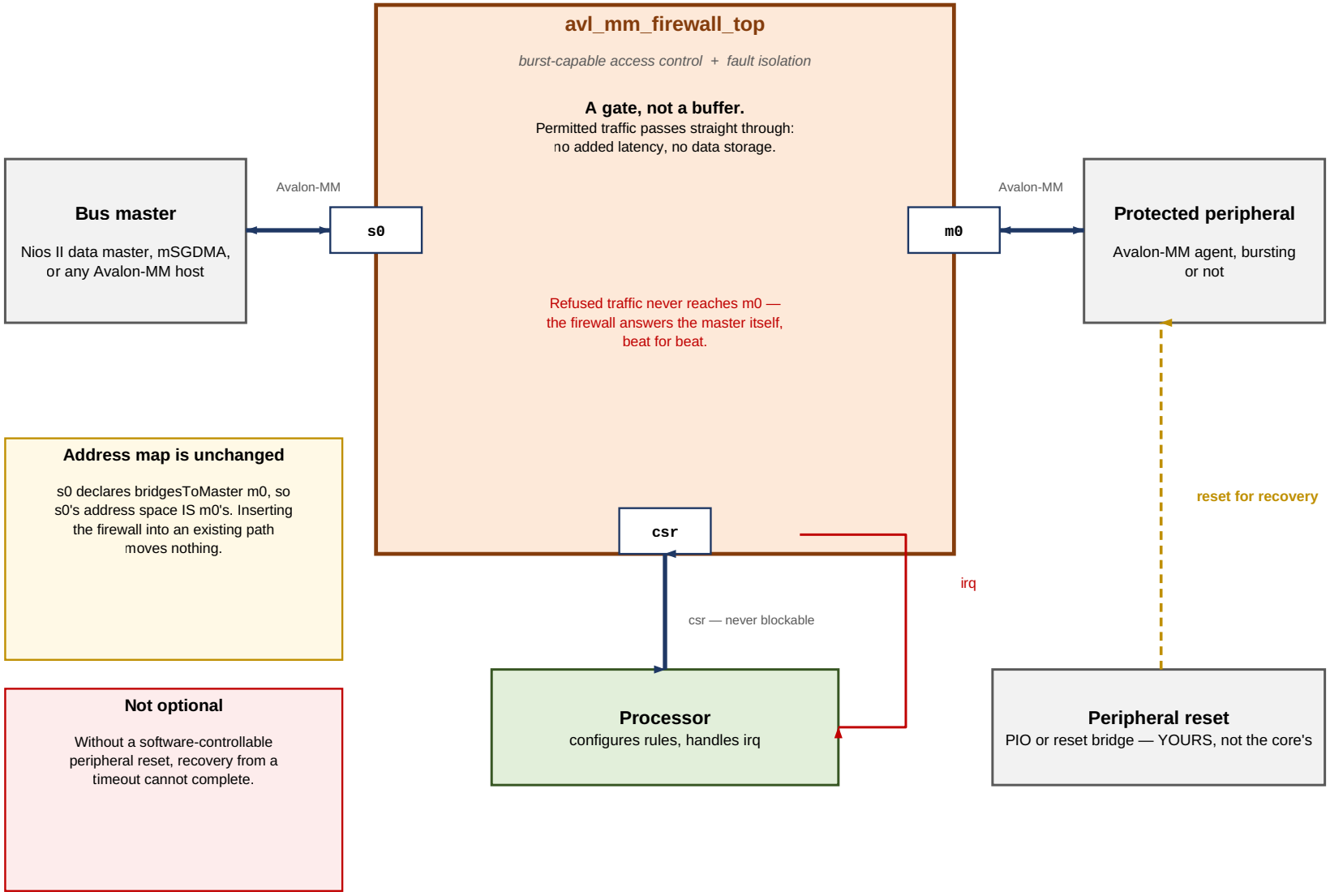


Figure 1. System context

The core is a bump in the wire. It has three Avalon-MM interfaces and one interrupt:

Interface	Role	Notes
`s0`	Agent, faces the master	Byte-addressed, bursting, pipelined reads
`m0`	Host, faces the protected peripheral	Mirrors `s0`
`csr`	Agent, configuration and status	32-bit, word-addressed, fixed read latency 1, never stalls
`irq`	Interrupt sender	Level, cleared by clearing the source

1.1 The address map does not move

`s0` declares `bridgesToMaster m0` in the Platform Designer component. That tells the interconnect that `s0`'s address space *is* `m0`'s: the firewall is transparent to address assignment, exactly like `altera_avalon_mm_bridge`.

The practical consequence is that inserting this core into an existing system changes nothing about where anything lives. Existing software keeps working, existing address assignments stay valid, and the firewall can be added late in a project without a re-layout of the memory map.

1.2 Why the control port is separate

`csr` is deliberately not reachable through `s0`. If configuration went through the policed path, then:

- a rule that accidentally excluded the firewall's own registers would lock software out permanently, and
- a wedged peripheral could block the very port needed to recover from it.

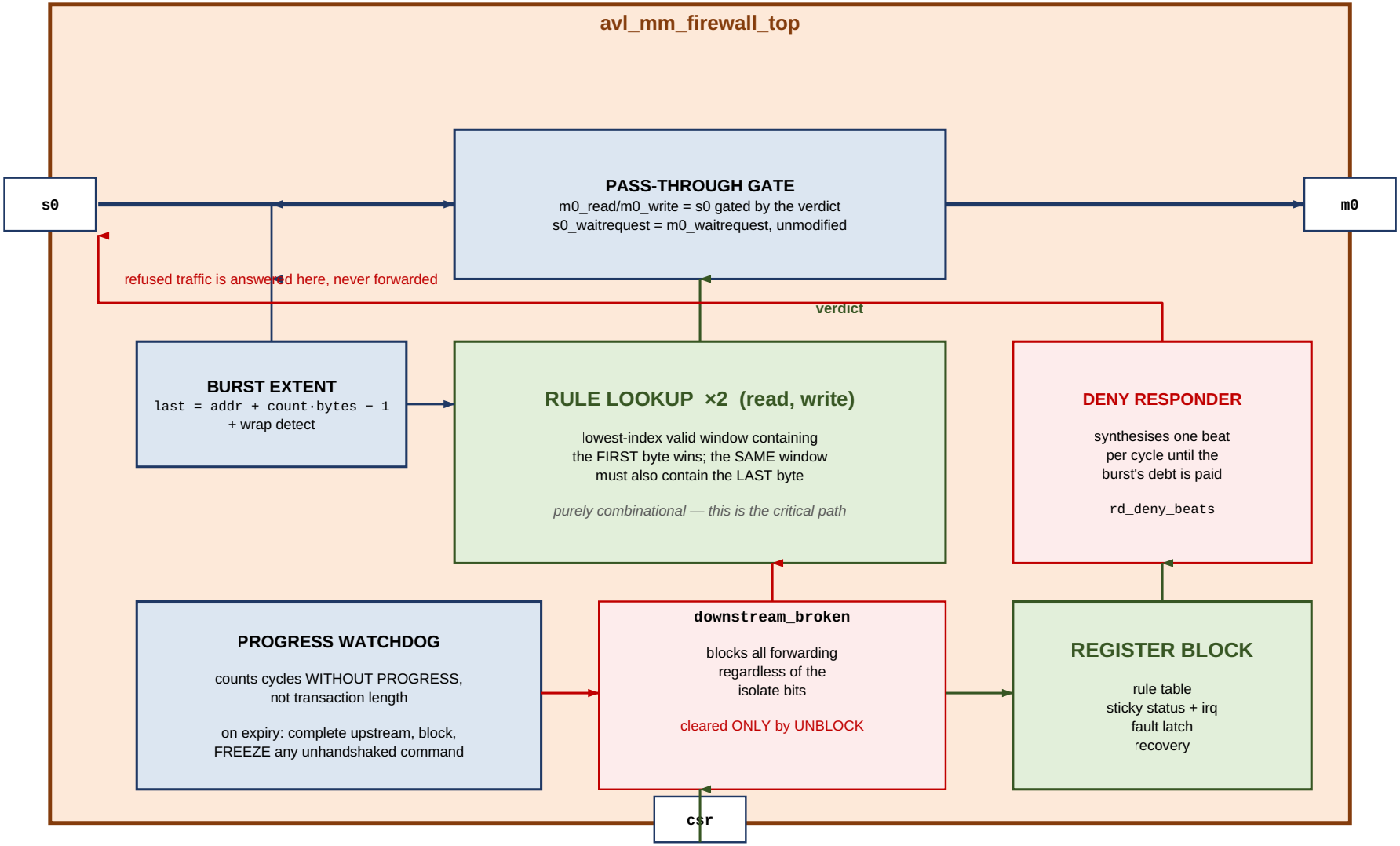
Keeping it separate also lets `csr` be the dullest possible Avalon-MM agent: word-addressed, fixed latency, no `waitrequest`, no outstanding-transaction state. There is nothing in it that can get stuck.

1.3 What the core does not own

The peripheral's reset is not driven by the firewall. That is a deliberate trade: owning it would demand a dedicated reset net per protected peripheral, which shared reset domains and hard IP frequently cannot provide.

The cost is that recovery from a timeout becomes a software sequence which *must* have a way to reset the peripheral. A system without one cannot complete recovery, and a single timeout takes the peripheral out of service until the whole system is reset. This is the single most important thing to get right at integration time.

2. Internal architecture



There is no data storage anywhere in the data path. Refused traffic is the only thing the core generates itself.

Figure 2. Internal architecture

2.1 A gate, not a buffer

The central decision is that permitted traffic is **gated, not captured**. `m0_read` and `m0_write` are `s0_read` and `s0_write` gated by the rule verdict; `s0_waitrequest` is `m0_waitrequest` unmodified; read data flows back untouched. There is no data storage anywhere in the data path.

This is what makes the core usable in front of a DMA engine. The alternative — capture, evaluate, re-drive — is what the AXI4-Lite sibling does, and it costs six cycles per transaction, which a burst cannot amortise because Platform Designer's burst adapter has already split it into single beats by then.

The price is paid in timing, not cycles: the rule lookup is combinational and sits between `s0_address` and both `m0_read` / `m0_write` and `s0_waitrequest`. It is the critical path, and it grows with `NUM_RULES`.

2.2 The blocks

Block	What it does
Burst extent	Computes `last = address + burstcount × bytes − 1` with one guard bit, so a burst wrapping the top of the address space is detected rather than aliasing
Rule lookup ×2	Two independent combinational lookups, one per channel, so read and write get an answer in the same cycle without contending
Pass-through gate	The gating itself
Deny responder	Owns `rd_deny_beats` : the debt the core owes a master whose burst it refused
Progress watchdog	Counts cycles without progress on a forwarded transaction
`downstream_broken`	The latch that walls the peripheral off. Set by the watchdog, cleared only by `RECOVERY.UNBLOCK`
Register block	Rule table, sticky status, interrupt, fault latch, recovery

2.3 Why the lookup takes two addresses

Each lookup port is given the **first and last byte** of the transaction, not just the address. It reports whether a valid window contains the first byte, and whether that *same* window also contains the last.

Checking only the start address produces a firewall that a bursting master walks straight through: begin one beat inside a permitted window, run for 128 beats, and you are reading whatever is mapped after it. For a core whose entire reason for existing is bursting masters, that would be the defect that matters most.

Two consequences follow from checking against the *same* window rather than the table as a whole:

- **Adjacent windows do not merge.** A burst crossing the boundary between two abutting permitted windows is refused. Permissions are per-window, and such a burst would have to satisfy both.
- `BURST_ALLOW` **is per-window.** "This peripheral cannot handle bursts" is a property of the peripheral, and a real system has both kinds behind one firewall.

2.4 Key internal signals

Signal	Meaning
`wr_beats_left`	Beats still expected in the current write burst. Non-zero means a burst is in progress and the verdict is latched
`wr_fwd`	The current write burst is being forwarded, as opposed to swallowed
`rd_fwd_beats`	Read beats owed by the peripheral. Also the orphan filter: `m0_readdatavalid` is forwarded only while this is non-zero
`rd_deny_beats`	Read beats the core itself owes the master
`wr_stuck` / `rd_stuck`	A command is frozen because its `waitrequest` never fell
`downstream_broken`	Forwarding is refused; only `UNBLOCK` clears it

3. How a transaction is judged

How one transaction is judged

Evaluated combinationally at the first beat, then held for the whole burst.

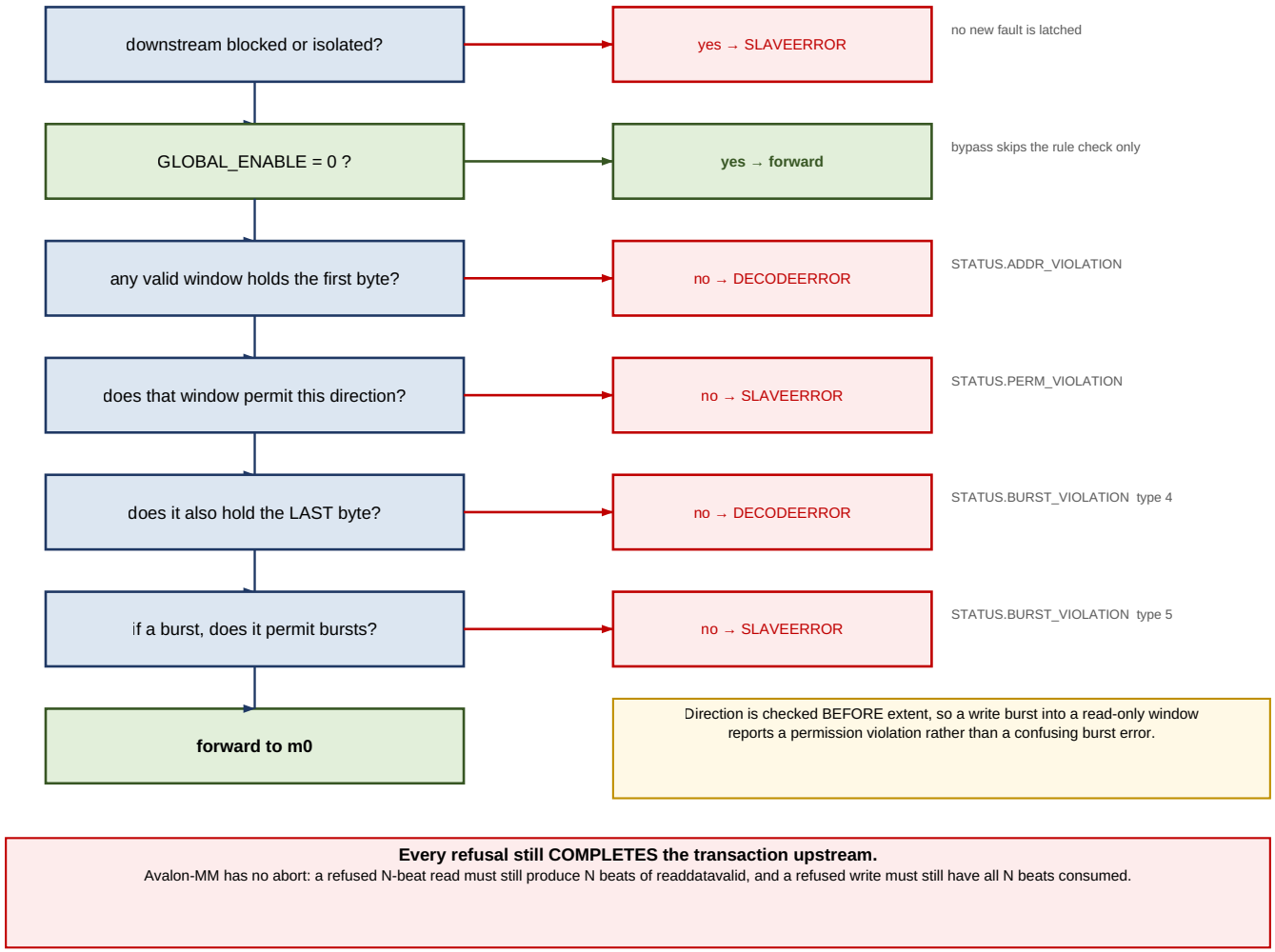


Figure 3. Verdict

The verdict is formed combinationally at the first beat and held for the whole burst. On beats 2..N of an Avalon-MM write burst the master presents only `writedata` — re-evaluating per beat would be evaluating garbage.

3.1 Order of checks

Two orderings are load-bearing.

Blocked before bypass. `CTRL.GLOBAL_ENABLE = 0` disables *access control*, not *fault isolation*. A peripheral already walled off after a timeout stays walled off in bypass mode. Access control and isolation are separate jobs, and a wedged peripheral is a bus-level hazard regardless of whether anyone is policing addresses.

Direction before extent. A write burst into a read-only window reports `PERM_VIOLATION`, not a burst error. The window is wrong for this access entirely; how far the burst would have run is a secondary detail and reporting it first would send whoever is debugging in the wrong direction.

3.2 Verdict, response and status

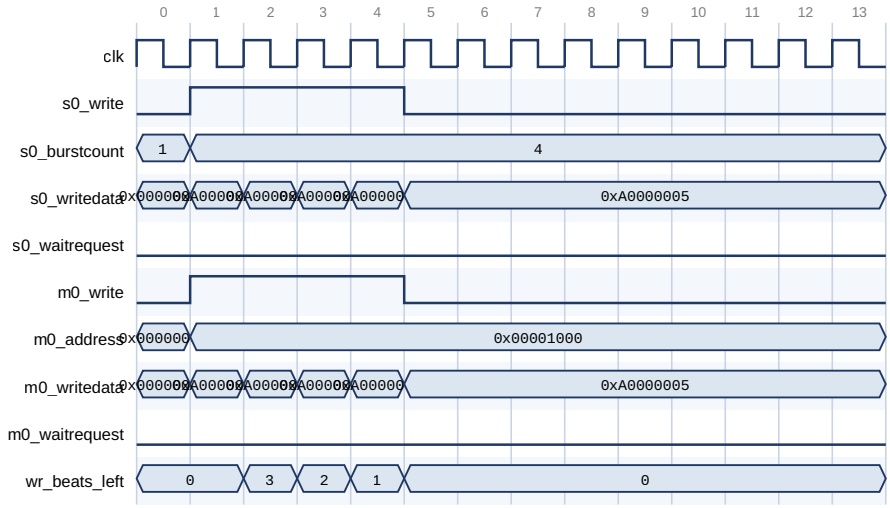
Verdict	Response	Sticky bit	`FAULT_INFO.TYPE`
Address in no valid window	`DECODEERROR`	`ADDR_VIOLATION`	1
Window forbids this direction	`SLAVEERROR`	`PERM_VIOLATION`	2
Downstream stopped progressing	`SLAVEERROR`	`TIMEOUT_ERROR`	3
Burst extends past the window	`DECODEERROR`	`BURST_VIOLATION`	4
Window forbids bursts	`SLAVEERROR`	`BURST_VIOLATION`	5
Refused because blocked	`SLAVEERROR`	*(none)*	*(none)*

`DECODEERROR` is used where the transaction reaches addresses that, as far as this window is concerned, are not there. `SLAVEERROR` is used where the address is real but the access is not allowed.

The last row is the subtle one. A rejection *while blocked* latches nothing, because the timeout that caused the block already latched a fault. Latching again on every subsequent rejected access would overwrite the `FAULT_ADDR` that actually diagnoses the problem, replacing it with whatever the driver happened to retry.

4. Bursts in motion

4.1 A permitted burst

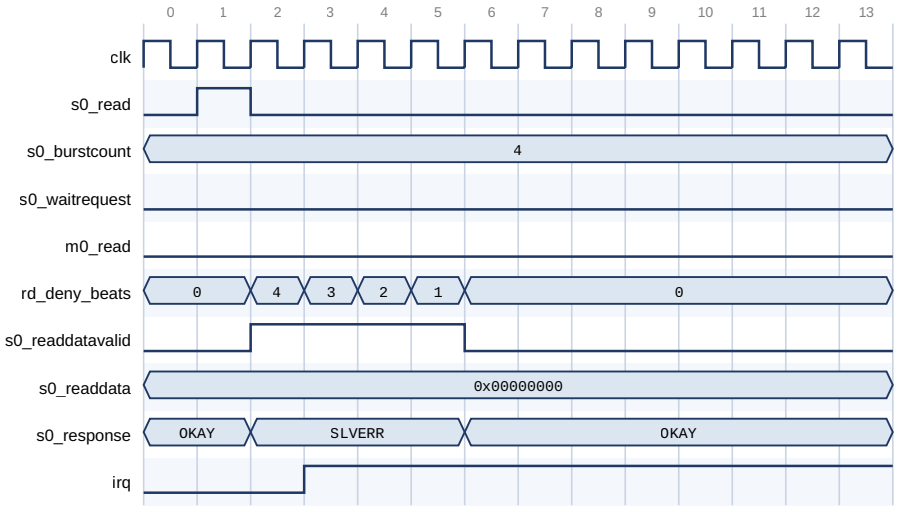


A 4-beat write burst into a window that permits writes and bursts.
m0 is s0 gated by the rule lookup: the same beats, the same cycle, no buffering. s0_waitrequest is m0_waitrequest unmodified, so the core adds no latency at all.
The address is meaningful only on the first beat; wr_beats_left is what holds the burst together, and the verdict formed at beat 1 governs all four.

Figure 4. Permitted write burst

Nothing happens to it. The beats appear on `m0` in the same cycles they appear on `s0`, and `s0_waitrequest` is `m0_waitrequest`. `wr_beats_left` is the only state involved — it holds the burst together so that the verdict formed at beat 1 governs beats 2 through N, whose addresses are not meaningful.

4.2 A refused burst



A 4-beat read burst into a write-only window. m0_read never asserts - the protected peripheral is not touched at all.
The command is ACCEPTED immediately (waitrequest stays low) rather than stalled: stalling it would move the hang from the peripheral into the firewall.
Avalon-MM has no way to refuse, so the core owes the master four beats and synthesises them itself. rd_deny_beats is that debt counting down. Data is driven to zero, not left stale.

Figure 5. Refused read burst

This is the case with no AXI analogue, and the reason a large part of the core exists.

Avalon-MM has no way to refuse a transaction. A refused 4-beat read must still produce four beats of `readdatavalid`, or the master waits forever. So "deny" does not mean "ignore" — the firewall becomes the responder:

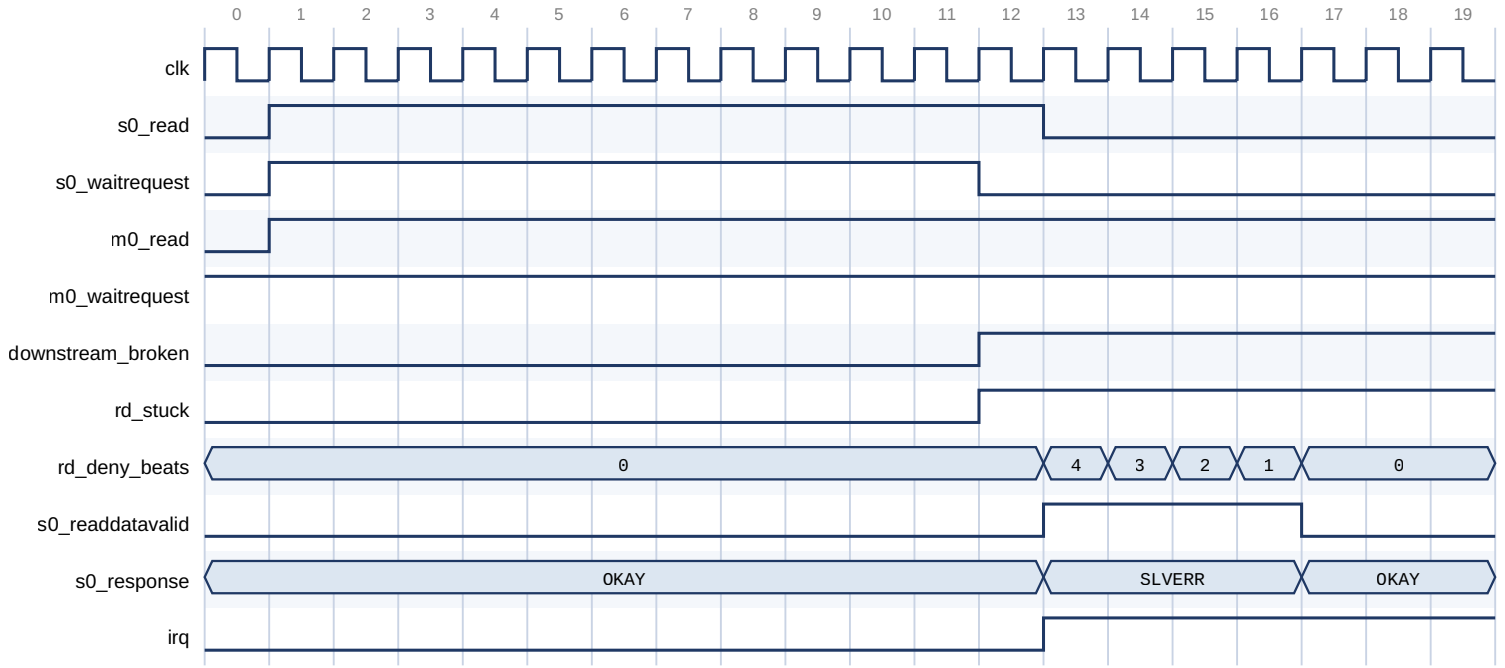
1. The command is **accepted immediately**, `waitrequest` low. It is never stalled on the rule check, because stalling would simply move the hang from the peripheral into the firewall.
2. `rd_deny_beats` is loaded with the burst length. That is the debt.
3. One beat per cycle is synthesised until the debt is paid, carrying the error response and **zero data, not stale data**.
4. `m0` is never touched. The protected peripheral does not see the transaction at all.

`irq` asserts on the way through, because the sticky status bit is set and its interrupt is enabled.

Only one refused read burst is in flight at a time, and no new read is accepted while one is draining. That is what satisfies Avalon-MM's in-order read response requirement without a reorder buffer. Refusals are errors, so the throughput cost is irrelevant; permitted reads pipeline freely.

5. Failure and recovery

5.1 The hang



The peripheral holds m0_waitrequest high forever. TIMEOUT_VALUE is 10 cycles in this capture; a real system uses thousands.

On expiry the core completes the transaction upstream itself - four SLAVEERROR beats - so the master never hangs, and latches downstream_broken.

m0_read STAYS ASSERTED. Avalon-MM forbids withdrawing a command before waitrequest falls; rd_stuck records that the core is holding one. Only RECOVERY.UNBLOCK may drop it.

Figure 6. Timeout

The Avalon-MM hang mode is `waitrequest` stuck high, or `readdatavalid` that never arrives. The watchdog counts cycles **without progress** — not transaction duration. Timing whole transactions would force `TIMEOUT_VALUE` to be scaled by the longest burst in the system, which makes it useless as a hang detector.

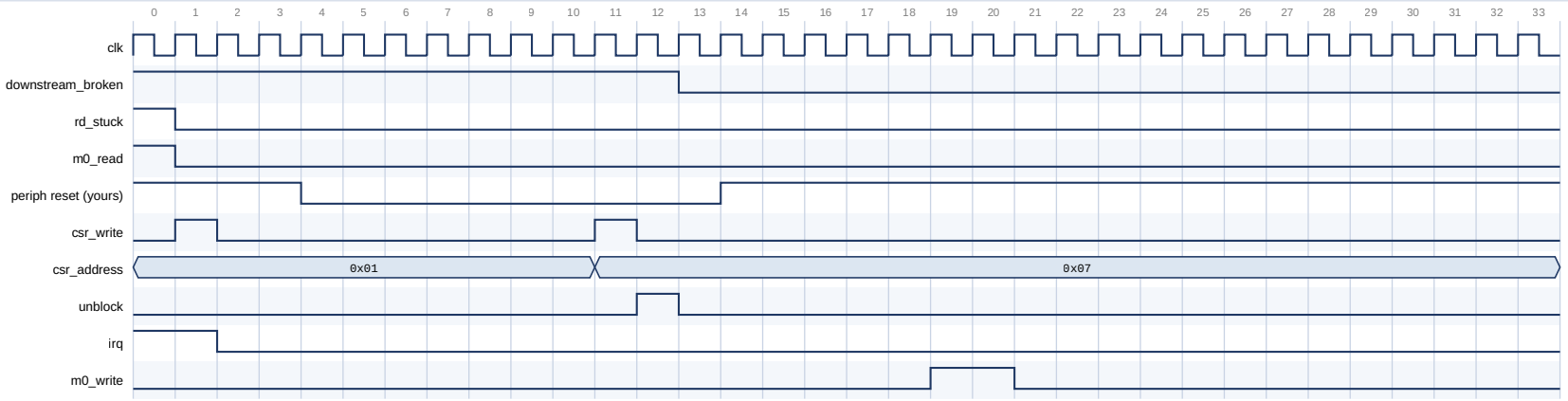
On expiry the core:

1. completes the transaction upstream itself, so the master never hangs;
2. latches `downstream_broken`;
3. **leaves `m0_read` asserted.**

Point 3 is the one to look at in the figure. Avalon-MM requires a host to hold `read` or `write` asserted until `waitrequest` deasserts. Withdrawing it can wedge the interconnect between the firewall and the peripheral, not merely the peripheral. So the core distinguishes two kinds of damage:

	Avoidable?	What the core does
A forwarded burst abandoned part-way, leaving the peripheral waiting for beats	No — the master must be released, and Avalon-MM has no burst-abort	Accepts it. This is why the peripheral reset is mandatory
A command withdrawn before its handshake completed	Yes	Never does it. `rd_stuck` / `wr_stuck` record that one is being held

5.2 The recovery



Word 0x01 (byte 0x04) acknowledges the fault and irq drops. The peripheral reset is driven by the system, not by the core.
Word 0x07 (byte 0x1C) pulses unblock WHILE THE PERIPHERAL IS STILL IN RESET. downstream_broken and rd_stuck clear together, and the held m0_read is withdrawn where the peripheral cannot observe it.
Doing this after releasing the reset would let the peripheral handshake the held command first - latching a transaction already reported as failed.
The write burst at the end shows forwarding has reopened.

Figure 7. Recovery

```
1. Stop issuing transactions to s0.
2. Write 1 to the sticky STATUS bits.
3. Poll STATUS until WR_BUSY and RD_BUSY clear - WITH A BOUND.
4. ASSERT the peripheral's reset and HOLD it.
5. Write RECOVERY.UNBLOCK - while the reset is still asserted.
6. Release the peripheral's reset.
7. Resume.
```

Steps 5 and 6 are in this order deliberately, and this is where the core deviates from AMD's published AXI Firewall flow, which resets and then unblocks. `UNBLOCK` is what withdraws the frozen command. If the peripheral is already out of reset when that write lands, it can complete the frozen command's handshake first — latching a transaction the firewall has already reported to the master as failed. Withdrawing it while the peripheral cannot see the bus removes that window, and costs nothing.

That ordering was not designed in. It came out of a coverage gap: the cover point `c_unblock_with_stuck_cmd` sat at zero hits with the reset-then-unblock sequence, because the freshly-released peripheral always handshaked the frozen command away before `UNBLOCK` arrived. A branch that refuses to be covered is sometimes telling you the sequence is wrong rather than that the stimulus is weak.

Bound the poll in step 3. `WR_BUSY` and `RD_BUSY` mean *the peripheral owes us something*, and a peripheral that accepted a command and then died owes it forever. An unbounded poll hangs exactly when recovery matters most.

Two further points visible in the figure:

- Acknowledging the fault (step 2) drops `irq` but does **not** clear `BLOCKED`. Clearing a fault must not accidentally restart traffic toward a peripheral nobody has reset.
- Traffic attempted while blocked is **answered with an error, not stalled**. There is no window in which the firewall quietly holds transactions until recovery finishes, so drivers need a retry path.

6. Register map and rule table

csr register map — byte offsets

The port is word-addressed in hardware; the interconnect does the divide-by-four, so software uses these offsets directly.

Offset	Name	Access	Purpose
0x00	CTRL	R/W	enable, auto-isolate, manual isolate
0x04	STATUS	R / W1C	four sticky faults, six live status bits
0x08	IRQ_ENABLE	R/W	one mask bit per sticky fault
0x0C	TIMEOUT_VALUE	R/W	cycles WITHOUT PROGRESS before giving up
0x10	FAULT_ADDR	R	start address of the latched fault
0x14	FAULT_INFO	R	direction, cause, burst length
0x18	CORE_INFO	R	generated parameters and version
0x1C	RECOVERY	W	UNBLOCK — the one authorised discard
0x40+i·0x10	RULE_BASE[i]	R/W	first byte of window i, inclusive
0x44+i·0x10	RULE_LIMIT[i]	R/W	last byte of window i, inclusive
0x48+i·0x10	RULE_PERM[i]	R/W	read / write / valid / burst

CTRL (0x00)

31:3	2	1	0
reserved	MANUAL_ISOLATE	AUTO_ISOLATE_EN reset 1	GLOBAL_ENABLE reset 1

STATUS (0x04) — bits 3:0 sticky, write 1 to clear

9:8	7:6	5	4	3	2:0
RD/WR_CMD_STUCK	RD/WR_BUSY	BLOCKED	ISOLATED	BURST_VIOLATION	TIMEOUT, PERM, ADDR

FAULT_INFO (0x14)

31:16	15:8	7:4	3:1	0
reserved	BURSTCOUNT saturating	reserved	1=ADDR 2=PERM 3=TIMEOUT 4=BURST_RANGE 5=BURST_DENIED	WAS_WRITE

RULE_PERM[i] (0x48 + i·0x10)

31:4	3	2	1	0
reserved	BURST_ALLOW	VALID	WRITE_ALLOW	READ_ALLOW

CORE_INFO (0x18)

31:16	15:13	12:8	7:0
version 0x0100 = v1.0	log2 bytes/beat	BURST_WIDTH	NUM_RULES

Figure 8. Register map and bit fields

6.1 Addressing

`csr` is word-addressed in hardware — Platform Designer's default for an Avalon-MM agent. Every offset quoted in the documentation is a **byte** offset, because that is what software passes to `IOWR_32DIRECT()`; the interconnect performs the divide-by-four. Nothing in a driver needs to scale anything.

6.2 The rule table

Each window is three registers at `0x40 + i × 0x10`:

Word	Register
+0x0	`RULE_BASE[i]` — first byte, inclusive
+0x4	`RULE_LIMIT[i]` — last byte, inclusive
+0x8	`RULE_PERM[i]` — read / write / valid / burst
+0xC	reserved, reads zero

The lowest-index valid window containing the start address wins. Windows need not be disjoint.

Because a window is described by three registers, updating a **live** window leaves a transient in which the base is new and the limit is still old — a window describing a range nobody intended, live on the bus, for a few cycles. Clear `VALID` first when reconfiguring an active window; default-deny then covers the gap. At initialisation this is unnecessary, since `VALID` is 0 out of reset.

6.3 Secure at reset

`CTRL` resets to `GLOBAL_ENABLE | AUTO_ISOLATE_EN` and the entire rule table resets invalid. The core therefore comes out of reset denying everything, with fault isolation armed. The interval between reset and the first configuration write is exactly when default-deny should be in force, which is why the driver never writes `CTRL` during initialisation.

6.4 Identifying the core from software

`CORE_INFO` at byte 0x18 describes the hardware as generated: `NUM_RULES` in [7:0], `BURST_WIDTH` in [12:8], log2 of the bytes per beat in [15:13], and the version in [31:16], which reads **0x0100** for v1.0.

A driver should check the version field before touching any other register. The register map is not self-describing beyond this word, and a driver writing v1.0 offsets into a future core would fail silently rather than loudly — the wrong direction to fail for a security peripheral.

6.5 The fault latch

`FAULT_ADDR` and `FAULT_INFO` are a single shared latch, updated by each new fault. `FAULT_ADDR` holds the **start address of the transaction**, not the address of the offending beat within a burst.

`FAULT_INFO.BURSTCOUNT` saturates at 255 rather than truncating: a 256-beat burst reading back as 0 would be worse than useless in a field whose whole purpose is to say how large the offending transfer was.

If a read fault and a write fault land in the same cycle, both sticky bits set correctly but the shared latch captures the write side. That is a documented, deterministic tie-break rather than a race.

Read the fault registers **before** acknowledging. Acknowledging reopens the core to traffic that can fault again immediately and overwrite them.